Line (GAC): L COMPACTOR OPERATION

Competency: L5 Operate Compaction Roller

Objectives

To be competent in this area, the individual must be able to understand and perform standard operating functions.

LEARNING TASKS	CONTENT
1. Plan work	• Pre-rolling considerations • Looking over grade for danger, obstructions and limitations • Determine number of Compaction Rollers/ type equipment • Ensure spray tanks are full • Make proper adjustments ○ Because following factors: weather, site conditions, grade, project densometer readings, vibration frequency ○ Compaction equipment speed – dictated by production
2. Test temperature	• Mix temperature
3. Test density	• Describe the use and purpose of nuclear densometer – quality control • Describe Troxler (manufacturer) • Understand technician certification • Determine roll pattern • Theoretical density • Frequency is job dependant – sometimes required after each pass
4. Maintain efficient patterns	• Pattern in both direction • Forward and reverse not expected to stop/park • Change direction smoothly and slowly • Breakdown rolling • Intermediate rolling • Finish rolling

Achievement Criteria

Performance
The learner will perform standard operating functions.
Learner will demonstrate safe operating practices during practical training and on the job.

Conditions
To complete the in-school practical lab requirement for the Compaction Roller, the learner must be able to demonstrate the following practical skills:

- Demonstrate the safe operation of equipment including observation of surroundings
- Perform pre-start checks, start-up/shut-down procedures and monitor performance of the equipment
- Perform daily maintenance tasks

Perform basic moves with equipment including:

- Move forward, stop, back up, stop (flat elevation)
- Apply park brake, lock-out transmission
- Shift transmission
- Operate vibratory Compaction Roller system
- Make a number of passes with Compaction Roller, demonstrating typical confined edge rolling pattern
- Make a number of passes with Compaction Roller, demonstrating longitudinal joint compaction pattern

To complete the jobsite requirement for the Compaction Roller, the learner must be able to demonstrate all of the above in-school practical lab requirements plus the following practical skills:

- Coordinate movement with the asphalt train (paver, breakdown Compaction Roller, intermediate Compaction Roller and finish Compaction Roller) to ensure that asphalt is compacted efficiently.
- Coordinate and communicate with other crew members effectively
- Plans work
  - Looks over the grade for dangers, obstructions and limitations
  - Has the ability to determine types of compaction rollers required for the job
  - Can make adjustments based on:
    - Weather
    - Site conditions, and
    - Grade
- Maintains efficient rolling patterns using Breakdown, Intermediate and Finish Rollers
- Able to determine the appropriate rolling pattern for the job.

Note
It is highly recommended that the apprentice complete the interactive training materials (CD's) created by Caterpillar

- Fundamentals of Asphalt Paving ($50)
- Fundamentals of Asphalt Compaction ($50)

Both CD's can be purchased through Caterpillar by email (Paving_Products_Online@cat.com)

Line (GAC): L COMPACTOR OPERATION

Competency: L6 Follow shut down procedures

Objectives

To be competent in this area, the individual must be able to perform correct shut-down procedures for the compactor.

LEARNING TASKS	CONTENT
1. Perform shut-down procedures	• Position that allows access in case of mechanical trouble at start-up • Safe parking position, set parking brake • Level position for checking of fluids • Cool down before shut down • Walk around, look for leaks, general check • Night switches/lockouts – batteries • Secure machine, locks, etc.
2. Clean equipment	• Cleaning the drum, etc.
3. Ensure efficient staging	• Foreman staging
4. Prepare for transport	• Transport driver responsible for moving
5. Secure machine for safety	• Public liability • Vandal proof take ancilary parts off • Park safety (can roll away)

Achievement Criteria

Performance	The learner will demonstrate correct shut-down procedures for the Compaction Roller.
Conditions	The learner will follow shutdown procedure as listed by the manufacturers guidelines.
Criteria	The learner will demonstrate a minimum of 80% efficiency using a checklist

Section 4

TRAINING PROVIDER STANDARDS

Facility Requirements

The following recommendations are offered to assist course instructors in establishing the most effective learning environment.

Classroom Area

- Sufficient space for 12-16 people to work comfortably at desks.

Shop Area

- N/A

Lab Requirements

- The facility requirements for the lab/practical portion of this course must allow participants access to an environment where they can view paving equipment at close quarters and to operate that equipment in a controlled setting for a reasonable amount of time. Suggested equipment time can be found within the Suggested Time Allocation Section, pages 14-16 of this document.

- Lab facilities will vary but should permit a simulated asphalt train. While it may not be reasonable or practical to use actual hot mix for training purposes, a simulated mix will give trainees a reasonable simulated experience.

Student Facilities

- Shall offer a safe and productive learning environment
- Meets applicable zoning bylaws for technical instruction and education
- Has access to sufficient land necessary to operate multiple pieces of equipment at the same time (suggested minimum of 6 acres)
- Conducts a safety review of the program's facility and equipment annually and meets applicable safety standards/regulations
- Suitable for the size of the class
- Temperature, noise, ventilation, light, and particulate control are maintained at appropriate levels

Instructor's Office Space

- Shall be appropriate for delivery of instruction at each level of training
- Storage space is functional and sufficient for instructional materials, supplies and equipment
- Instructor work stations are adequate and appropriately equipped
- Work Stations are adequate and appropriately equipped
- Facilities have adequate floor area and ceiling height
- Classroom theory and labs are heavily supported with a PowerPoint presentation for each section of the manuals. Classroom projection equipment with MS PowerPoint capabilities will be required.

Tools and Equipment

Shop Equipment

Required

- Compaction Roller (one for breakdown work, one for intermediate work, and one for finish work)
- Milling machine
- Paver
- Screed

Recommended

- N/A

Shop (Facility) Tools

Standard Tools

- Lutes
- Rakes
- Shovels
- Wheel barrows
- Various hand tools
- Tiger torch
- Measuring devices
- Densometer
- Propane tanks and hoses

Specialty Tools

- N/A

Student Equipment (supplied by school)

Required

- Does not apply

Recommended

- N/A

Student Tools (supplied by student)

Required

- PPE including Hi-Vis Vest, CSA approved steel toe boots, hard hat, gloves and hearing protection

Recommended

- N/A

Reference Materials

Required Reference Materials

- Instructor Guide, PowerPoint Slides, along with Common Core, Raker, Compaction Roller, Screed, Paver and Milling Machine manuals.
- 2006 Design Build Specifications
    - http://www2.gov.bc.ca/assets/gov/driving-and-transportation/transportation-infrastructure/engineering-standards-and-guidelines/highway-specifications/design_build_standard_specs_2006.pdf
- The Asphalt Handbook MS-4 7th Edition
- Interactive Training Materials (CD) through Caterpillar
    - Fundamentals of Asphalt Paving
    - Fundamentals of Asphalt Compaction
        - Both CD's can be purchased through Caterpillar by email (Paving_Products_Online@cat.com)
- WorkSafeBC on-line resources
    - http://www.worksafebc.com/

Recommended Resources

- http://www.asphaltinstitute.org/
- http://www.forconstructionpros.com/print/Pavement/Cover-Story/Buying-Your-First-Paver/3FCP4103
- http://www.paversearch.com/glossary-pavers.htm
- http://www.madehow.com/Volume-3/Asphalt-Paver.html
- http://www.grainger.com/Grainger/items/3YU79
- http://www2.worksafebc.com/Portals/Construction/Prevention-RoadConstruction.asp?ReportID=23946
- http://safety.cat.com/cda/layout?m=128061&x=7
- http://www.hotmix.org/allaboutasphalt.php
- https://www.ec.gc.ca/cov-voc/default.asp?lang=En&n=05CE2B41-1

Instructor Requirements

Occupation Qualification

The instructor must possess:

- Asphalt Paving/Laydown Technician Certificate of Qualification with endorsements for the pieces of equipment being taught.

Work Experience

A minimum of 5 years experience working in the industry as a journeyperson.

Instructional Experience and Education

- N/A

Field Training with an Employer

The apprentice is expected to gain further practical skills in the field and both employer and apprentice are expected to keep track of equipment hours and competencies acheived. The apprentice's supervisor will be permitted to sign off on any hours that the apprentice has gained for the program. Competency sign off for equipment skills will require sign off by a person holding a Certificate of Qualification with an endorsement for that machine or Supervision and Sign-Off Authority granted by SkilledTradesBC.